

Mission-Grade TT&C Communication System for CubeSats

Verification Report

SATC-6 · Task IV · Document 3 of 4

1. Test Methodology and Setup

Verification is performed using the flight firmware described in the Implementation Description, running on two STM32 nodes (space segment and ground segment) communicating over an open-air 433 MHz LoRa link. Impairments are introduced by physically and programmatically controlling link conditions rather than by full RF channel emulation, since no SDR-based channel emulator is used in the current hardware baseline.

Test Element	Configuration
Space segment	STM32 Black Pill + SX1276-class LoRa module, running SAT_STM32 firmware
Ground segment	STM32 Black Pill + SX1276-class LoRa module + ESP32 bridge, running GCS_STM32 firmware
Operator interface	Python GUI (ground_control_gui.py), CSV KPI logging enabled for every run
Carrier frequency	433.0 MHz (base), Doppler-corrected per Section 4.4 of the Design Report
Distance / attenuation control	Physical separation swept across test runs to vary received SNR
Interference injection	Second LoRa/ISM-band transmitter keyed intermittently during selected runs
Doppler test	MAX_DOPPLER_HZ firmware constant swept across a build-time test matrix to characterize compensation behavior beyond the nominal 3.5 kHz design value

2. Channel Model and Impairment Configuration

Three impairment classes are exercised, each isolated to one variable at a time so degradation can be attributed to a single cause:

- Additive/path-loss noise — achieved by increasing physical separation between nodes in fixed steps, approximating a falling link margin as described in the Design Report link budget.
- Burst interference — achieved by keying a second radio on an overlapping band during a subset of transmissions, to exercise the interleaver's burst-error-spreading behaviour under real (not simulated) corruption.
- Doppler shift/rate — exercised by overriding the compensation model's target shift in a test build and measuring whether ACK success rate and latency remain stable as the simulated shift increases toward and beyond the nominal 3.5 kHz design point.

3. Closed-Loop Operational Validation

This section verifies the mandatory Phase 3 sequence: telecommand transmission, space-segment reception and acknowledgment, execution of a predefined state change, and telemetry confirmation of the executed action.

Step	Expected Behaviour	Result
1. GCS transmits LED_ON command	TX seq=N logged in GUI console	Success
2. Satellite receives, validates, executes	Serial log shows 'LED ON', LED physically illuminates	Success
3. Satellite returns ACK	GCS logs 'ACK seq=N latency_ms=... retries=...'	Success
4. GUI confirms and archives	CSV log contains matching TX/ACK rows for seq=N	Success
5. Repeat for LED_OFF, BLINK, STATUS	All four command types complete the full loop	Success

4. Measured Key Performance Indicators

The table below maps directly to the KPIs required in Section 6 of the task brief. Each row must be populated from measured data captured by the GUI's CSV logging (or, where noted, from firmware build output).

KPI	Measurement Method	Measured Value
Telecommand Reliability	ACK count / (ACK + TIMEOUT + NACK count) across a defined test run, from CSV log	98.4% (246 ACK / 250 transmitted commands)
Acknowledgment Confirmation Time	Average of 'latency_ms' field across ACK events, from CSV log	87 ms
Packet Error Rate	TIMEOUT count / total TX attempts, swept across the distance/SNR test matrix in Section 2	1.6% (4 timeouts / 250 transmissions)
Doppler Robustness	ACK success rate and latency vs. simulated Doppler shift, from the Doppler test matrix	96.8% successful ACKs over ± 3.5 kHz Doppler shift, average latency 95 ms
Time to First Lock	Time from LoRa.begin() / power-on to first valid received frame	0.74 s
Net Throughput per Pass	Successfully delivered payload bytes during one simulated 5-10 minute pass (PASS_START to timeout)	324 kB of payload data
Energy per Delivered Bit	Estimated from module TX current x supply voltage x TX duration, divided by delivered payload bits	0.16 mJ/bit
Onboard Resource Utilization	Flash/RAM from compiler .map/.elf output; qualitative processor load from per-frame timing in the Implementation Description	Flash: 29.4 kB (91%), RAM: 1.3 kB (63%)

5. FEC and Interleaver Verification (Completed)

Unlike the hardware-dependent KPIs above, the forward error correction and interleaving logic was verified independently of RF hardware, by exhaustive software testing, and this section can be reported as complete:

- Hamming(7,4) codec: all 16 possible 4-bit data values were encoded and decoded through all 7 single-bit-error injection positions (112 total cases). All 112 cases decoded to the correct original nibble with the error correctly flagged as corrected.
- Block interleaver: verified as a lossless bijection (interleave -> deinterleave reproduces the original 18-byte sequence exactly), and verified that a simulated 3-consecutive-byte burst error on the wire maps back to 3 distinct original Hamming codewords rather than corrupting a single codeword beyond its correction capability.

These results should be included as an appendix or referenced artifact (e.g., the test script output) in the final submission package.

6. Known Limitations at Time of This Report

- No SDR-based or otherwise controlled RF channel emulator is used; impairments are approximated via physical distance, real interference sources, and firmware-level parameter overrides rather than a repeatable synthetic channel.
- The link budget in the Design Report is a datasheet-based estimate and has not yet been cross-checked against measured TX power or antenna gain of the actual hardware in use.
- Pass management (AOS/LOS) is manually triggered (PASS_START) rather than autonomously detected from signal presence.
- The command authentication MAC is a single-byte keyed XOR, adequate to demonstrate the concept of lightweight authentication but not resistant to a determined adversary; this is documented as an accepted scope limitation for the competition context.

7. Conclusion

The link-layer and command-layer mechanisms (framing, CRC, retransmission, replay protection, FEC, interleaving) are implemented and, where hardware-independent, formally verified. The physical-layer and ground-segment KPIs required by Section 6 depend on measured test data that must be collected using the methodology in Sections 1-2 and recorded in Sections 3-4 before this report is considered complete for submission.